0215

National University of Computer and Emerging Sciences

Lahore Campus

Database Systems (CS2005) Sessional-2 Exam

Date: Mon, 06 Apr 2026

Course Instructor(s)

IR, ZA, MM, MN, AA

Total Time (Hrs.): 1

Total Marks: 20

Total Questions: 3

19

24L-0978 Roll No

4E Section

Student Signature

Instructions: Attempt all the questions in the space provided. Extra sheets will not be checked.

CLO # 3: Develop a normalized relational design to remove anomalies in a set of relations.

Q. No 1: Consider the relation schema R(A, B, C, D, E), with FDs F= {AB→CD, A→C, C→D, AB→E, E→A}. Find a minimal cover of *F* (i.e. F_c). [4]

1) Normalize R.H.S into single attibute

=> {AB-C, AB-D, A-C, C-D, AB-E, E-A} = FDS

② For Every Fd, check ifor the multi-altibut L.H.S Fds, if any attibute is redudant

(i) in AB-C, check only $A^+ = \{A, C, D\}$ C is present so B is redundant

AB-C => A-C

(ii) same for Rest AB-D, $A^+ = \{A, C, D\}$ present, so B redundant

AB-D = A-D

(iii) AB-E

$A^+ = \{A, C, D\}$ not in either

$B^+ = \{B\}$ So both AB required.

③ Remove duplicate / Redudant Fds.

① A-C, ② AB-E, ③ A-D, ④ C-D, ⑤ E-A

(we ignore the Fd at hand & check closure)

4

* check redundancy:

for ① $A^+ = \{A, D\}$ so A-C not redundant

for ③ $A^+ = \{A, C, D\}$, so A-D redundant

for ② $AB^+ = \{A B C D\}$, not giving E so not redundant

for ④ $C^+ = \{C\}$ = not redundant

for ⑤ $E^+ = \{E\}$ = not redundant

Final MC = {A→C, AB→E, C→D, E→A}.

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CLO # 3: Develop a normalized relational design to remove anomalies in a set of relations.

Q. No 2: Check whether the following decompositions of relations are dependency preserving and lossless. Show formal proof of your answer. [6]

a. R(A, B, C, D, E), FDs: F= {A→B, B→C, C→D, D→E}, Decomposition: R1(A, B, C) & R2(C, D, E).

b. S(A, B, C, D), FDs: F= {A→B, B→D, AC→D}, Decomposition: S1(A, B), S2(B, D) & S3(A, C).

a) lossless check

R1 ∩ R2 gives (C)

So take $C^+ = \{C, D, E\}$

= Candidate key of R2 = joinable

R1 ⋈ R2 = (A B C D E) = lossless

decomposition test

R(ABC)

$A^+ = \{B, C\}$

$B^+ = \{C\}$

$C^+ = \{\}$

Fds = {A-B, B-C}

R(C, D, E)

$C^+ = \{D, E\}$ 5-5

$D^+ = \{E\}$

$E^+ = \{\}$

Fds = {C->DE, D-E}

G = {A-B, B-C, C->DE, D-E}

All Fds can be validated to cover F and vice versa as the Fds upon joiney R1 & R2 are same where C-E is an extra indirect Fd.

No need for closure checking for each Fd attribute to check cover as its directly visible

F ⊆ G & F ⊇ G

F = G

= Decomposition test ~~failed~~ Passed

b) lossless check

$S1 \cap S2 = \{B\}$

$B^+ = \{B, D\} = S2$ = join

S1 ⋈ S2 = (A B D)

(S1 ⋈ S2) ⋈ S3 = (A)

$A^+ = \{A, B, D\}$

= CK of (S1 ⋈ S2)

= lossless = (A B C D)

decomposition test

S(A, B)

$A^+ = \{B\}$

$B^+ = \{\}$

Fd = A-B

S(BD)

$B^+ = \{D\}$

$D^+ = \{\}$

Fd = B-D

S(AC)

$A^+ = \{\}$

$C^+ = \{\}$

Fd = null

G = {A-B, B-D}

* Does F cover G?

Yes visibly A-B & B-D are stated in F so yes.

* Does G cover F?

G cover A-B, B-D in F as stated but does not cover AC-D which is given proof in null set of FD by S(AC).

decomposition test Failed

F ≠ G.

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CLO # 3: Develop a normalized relational design to remove anomalies in a set of relations.

Q. No 3: Identify the best normal form of the following relations. Justify your answer. If the given relation is not in BCNF, decompose it into a set of BCNF relations and show your steps. [10]

a. Consider the relation R(A, B, C, D), with FDs F= {AB→C, C→A, C→D}.

	AB−C	C−A	C−D
BCNF	✓	✗	✗
3NF	✓	✓	✗
2NF	✓	✓	✗

Keys

$AB^+ = \{ACDB\}$
$CB^+ = \{CBAD\}$ ✓

PA = A, B, C
NP = {D}

C−D fails partial dependency test
C = non prime attribute gives a non prime.
{ partial subset of CK }.

Highest NF = 1NF.

causing violation
$C^+ = \{C, A, D\}$

taking R_1 ~~(C, A)~~ ~~R_2 (C, A)~~

= R_1 (A, C, D) , R2 (A B C)

$A^+ = \{\}$
$C^+ = \{A, D\}$
$D^+ = \{\}$
Fd = {C → AD}
CK = C
= BCNF, L.H.S = CK.

$B^+ = \{\}$
$C^+ = \{A, D\}$
$A^+ = \{\}$
~~Fds = {C → AD}~~
~~CK = C~~
CK = AB
$AB^+ = \{A, B, C\}$
Fds = {AB − C, C − A}
BCNF NOT BCNF

4.5

Final Relation

= R_1(ACD) R_2(~~CA~~ ~~R_3(AB)~~ BC)

U = (A B C D) = Lossless

Fds Lost = AB − C

$C^+ = \{C, A\}$
R(C A)
Fd =
Fd = {C − A}
CK = C
= BCNF

(A B)
A − {}
B −
AB − {C}
− {}

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b. Consider the relation R(A, B, C, D, E), with FDs F= {A→B, B→C, C→A, CD→E}.

keys

	A - B	B - C	C - A	CD - E
BCNF	✗	✗	✗	✓
3NF	✓	✓	✓	✓

= CD
BD
AD

Highest normal Form = 3NF
given R.H.S = Prime attibute

Now to BCNF

~~A→B~~ $A^+ = \{A, B, C\}$

R(ABC)

Fds A - BC Bad ✓
B - C ✗
C - A ✗

ck = A

$B^+ = \{B, C, A\}$

(A B C)

A -> BC
A = CK
BCNF

R(ADE)

$A^+ = \{B, C, A\}$
$D = \{\}$
$E = \{\}$
Fd = none

(A)

A - {}
no fd

Final Relation in BCNF

= R(ABC) R(ADE)
~~R(ADE)~~ R(A)

= in BCNF.